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DevOps and Software Engineering

Job-ready skills | Career-focused | Shareable credential

Introductory | Self-paced | 52 weeks | Earn a certificate

15 courses , 1 year 1 month

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Program overview

Software Engineering professionals, especially those with DevOps skills, are in high demand! According to a recent GitLab report, the need for DevOps skills is expected to grow 122% over the next five years, making it one of the fastest growing markets in the workforce. Additionally, Glassdoor's salary report shows the average total pay for a DevOps Engineer in the US is $121,000. There has never been a better time to start a DevOps career path.

This self-paced certificate program of online courses, built for beginners, will equip you with the key DevOps concepts and technical know-how to build your Software Development skills and knowledge with DevOps practices, tools, and technologies. By the end of this program, you will be prepared for an entry-level role in Software Engineering with an organization of DevOps practitioners.

You will develop skill sets in a variety of DevOps philosophies, fundamentals, and methodologies, including Agile Development, Scrum Methodology, Cloud Native Architecture, Behavior and Test-Driven Development (BDD and TDD), and Zero Downtime Deployments. The program also touches on data science, cloud computing, programming languages, machine learning, continuous delivery, IBM cloud, and agile software development for DevOps professionals.

Guided by experts at IBM, you will learn how to:

- program with the Python language and Linux shell scripts,
- create projects on GitHub,
- containerize and orchestrate your applications using Docker, Kubernetes & OpenShift,
- compose applications with microservices,
- employ serverless technologies,
- perform continuous integration and delivery (CI/CD),
- develop test cases,
- ensure your code is secure,
- and monitor & troubleshoot your cloud deployments.

Labs and projects in this program are designed to equip you with job-ready hands-on skills to help you launch a new career in an in-demand field.

This software engineering professional certificate is suitable for those with no or some programming experience and those with or without college degrees.

Show less

There are 15 courses in this program

Software Engineering Basics for Everyone

10 - 14 hours per week, for 6 weeks

Build the foundation for an in-demand Software Engineering related career. Explore fundamental software engineering principles, processes, architecture, tools, and stacks. Hear from software experts discuss what it takes to be successful.

View course

DevOps Basics for Everyone

3 - 4 hours per week, for 3 weeks
This course is intended to help launch your career as a DevOps Professional. In this course you will learn to think, work, organize and measure DevOps - skills that you can use to build better products.

View course

Introduction to Cloud Computing

2 - 4 hours per week, for 3 weeks
Jumpstart your cloud expertise and explore offerings of AWS, Google, and IBM; delve into IaaS, PaaS, SaaS models, and cloud security. Experience IBM Cloud app deployment. Perfect for executives, students, and budding professionals.

View course

Agile and Scrum Fundamentals

3 - 4 hours per week, for 3 weeks
This course will benefit anyone who wants to get started with working the Agile way. It is particularly suitable for IT practitioners such as software developers, development managers, project managers, product managers, and executives.

View course

Linux Commands & Shell Scripting

3 - 4 hours per week, for 1 week
This mini-course describes shell commands and how to use the advanced features of the Bash shell to automate complicated database tasks. For those not familiar with shell scripting, this course provides an overview of common Linux Shell Commands and shell scripting basics.

View course

Git and GitHub Basics

6 - 8 hours per week, for 3 weeks
Gain the skills to work with Version Control Systems Git and GitHub. Explore key concepts, including branching and repositories; practice forking, cloning, and merging workflows and build your portfolio as your final project.

View course

Python Basics for Data Science

4 - 10 hours per week, for 3 weeks
This Python course provides a beginner-friendly introduction to Python for Data Science. Practice through lab exercises, and you'll be ready to create your first Python scripts on your own!

View course

Python for AI & Development Project

3 - 4 hours per week, for 1 week
This mini-course introduces you to the process for unit testing Python code, including building and running unit tests and packaging the Python application for distribution.

View course

Introduction to Containers, Kubernetes and OpenShift

2 - 3 hours per week, for 5 weeks
This course introduces you to containers and its emerging ecosystem of related technologies such as Docker, Kubernetes, OpenShift, and Istio. You will not only understand the concepts but also practice hands-on in the cloud.

View course

Microservices and Serverless

6 - 8 hours per week, for 2 weeks
Design, develop, deploy, manage and secure applications and solutions on public, private or hybrid cloud platforms

View course

Test and Behavior Driven Development (TDD/BDD)

14 - 16 hours per week, for 4 weeks
This course provides the fundamentals of Test Driven Development (TDD) and Behavior Driven Development (BDD), and why it’s essential for building quality applications and preventing bugs, reducing development costs and improving performance.

View course

Continuous Integration and Delivery (CI/CD)

9 - 11 hours per week, for 5 weeks
This course is an introduction to Continuous Integration and Continuous Delivery (CI/CD), a DevOps best practice to ensure code quality is achieved. It includes multiple hands-on labs using real-world tools to practice your CI/CD skills.

View course

Application Security for Developers

8 - 10 hours per week, for 5 weeks
Learn to identify security vulnerabilities in applications and implement secure code practices to prevent events like data breaches and leaks. Become familiar with DevSecOps practices, and SAST for identifying security flaws.

View course

Monitoring and Observability for Application Developers

8 - 10 hours per week, for 3 weeks
Monitoring & Observability are must have skills for Developers and DevOps professionals. This course provides a comprehensive overview of these concepts and teaches you the hands-on skills to employ monitoring, observability, and logging for your application.

View course

DevOps and Software Engineering Capstone Project

8 - 10 hours per week, for 5 weeks
In this DevOps Capstone Project, you'll demonstrate your skills and knowledge gained throughout this program with a real-world inspired hands-on project developing and deploying an application using CI/CD to showcase in your portfolio.

View course

What you'll learn

- Build your DevOps skillset, practice Agile philosophy, and Scrum methodology - essential to succeed in the DevOps field.
- Develop applications using Python language and various programming constructs and logic, including functions, REST APIs, and libraries.
- Create applications composed of microservices and deploy using containers (e.g., Docker, Kubernetes, and OpenShift) and serverless technologies.
- Employ tools for automation, continuous integration (CI) and continuous deployment (CD) including Chef, Puppet, GitHub Actions, Tekton, and Travis.

Meet your instructors

Rav Ahuja

Global Program Director at IBM Skills Network

Lin Joyner

Senior Instructional Designer & Content Developer at Skill-Up Technologies

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Industry insights

- **According to a recent GitLab report, the need for DevOps skills is expected to grow 122% over the next five years, making it one of the fastest growing markets in the workforce.**
- **Salary.com published the following in August 2022: The average DevOps Engineer salary in the United States is $124,566, but the salary range typically falls between $115,237 and $135,056.**

Why take this program?

Start learning a new skill today.

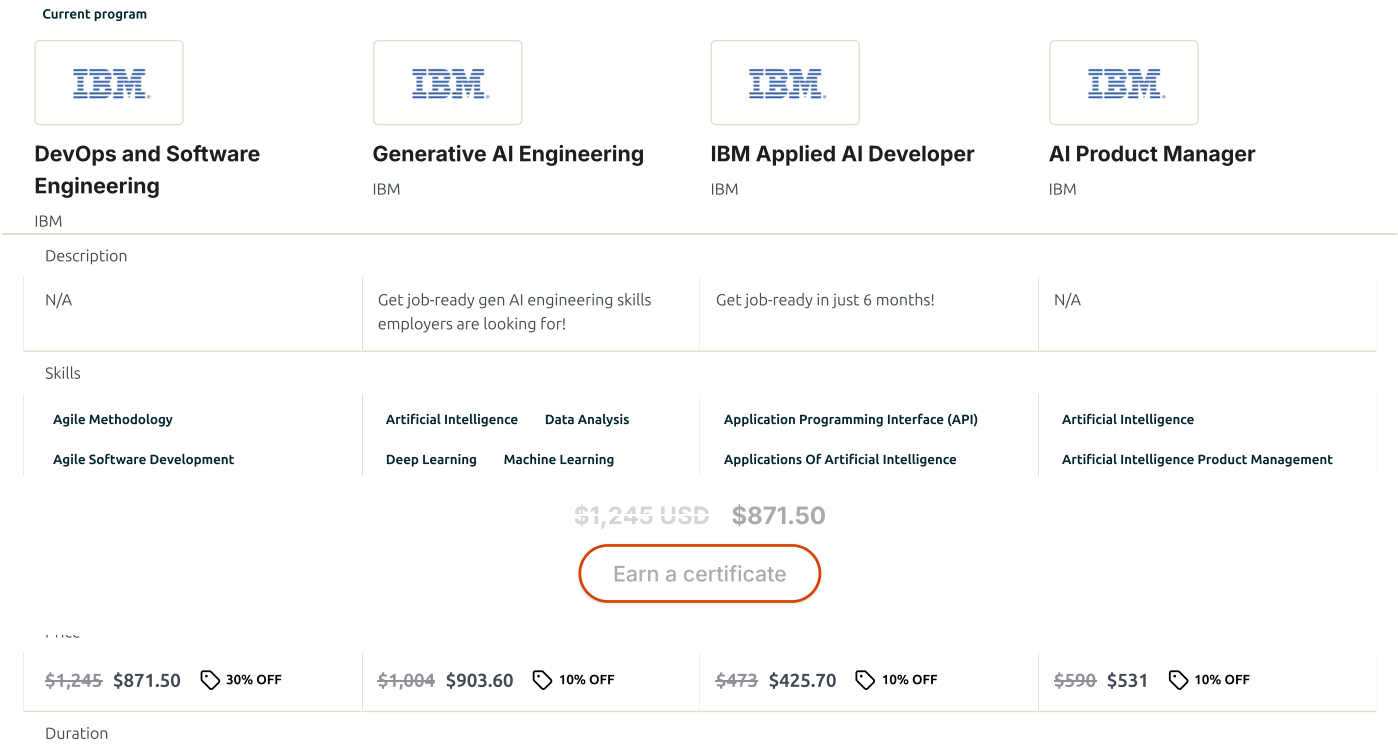
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13 months	13 months	6 months	10 months
Skill level			
Intermediate	Intermediate	Intermediate	Intermediate
Flexibility			
Self-paced	Self-paced	Self-paced	Self-paced
Earn a certificate	Learn more	Learn more	Learn more

Plan for a career in data science

Average 2024 salary

$112,590

Estimated job openings

23,400

Key skills: Cloud Computing | Github | Data Science | Unix Shell | Docker (Software) | Continuous Integration

This information is US-specific and can be found at the Bureau of Labor and Statistics as of 2025 .

er learners have to say

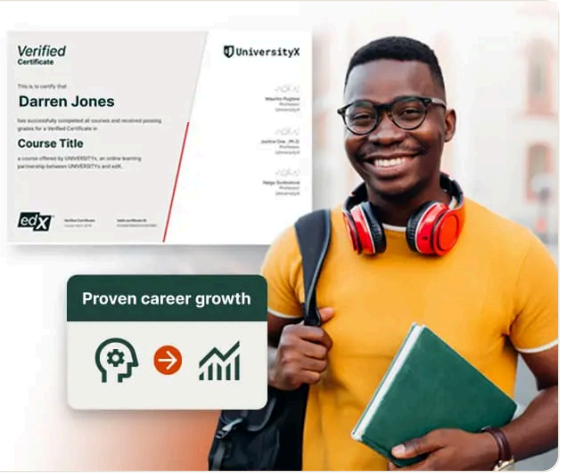
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